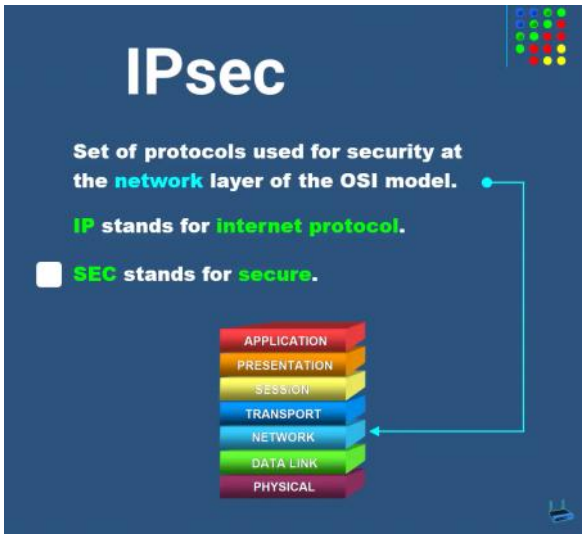
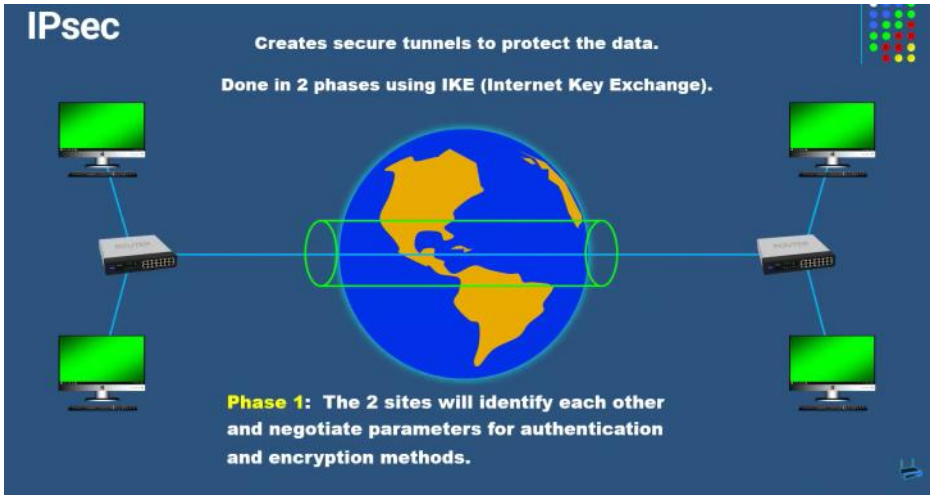
IPsec

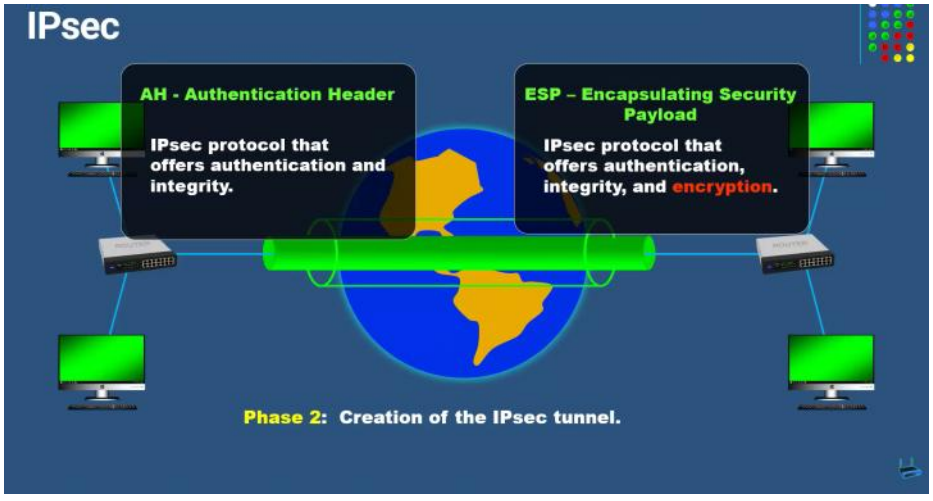
Wednesday, April 8, 2026 10:43 AM

IPsec (Internet Protocol Security) is a set of protocols designed to secure network-layer communications. It protects data sent between two points by adding encryption and authentication, making it the standard technology for establishing VPNs (Virtual Private Networks).



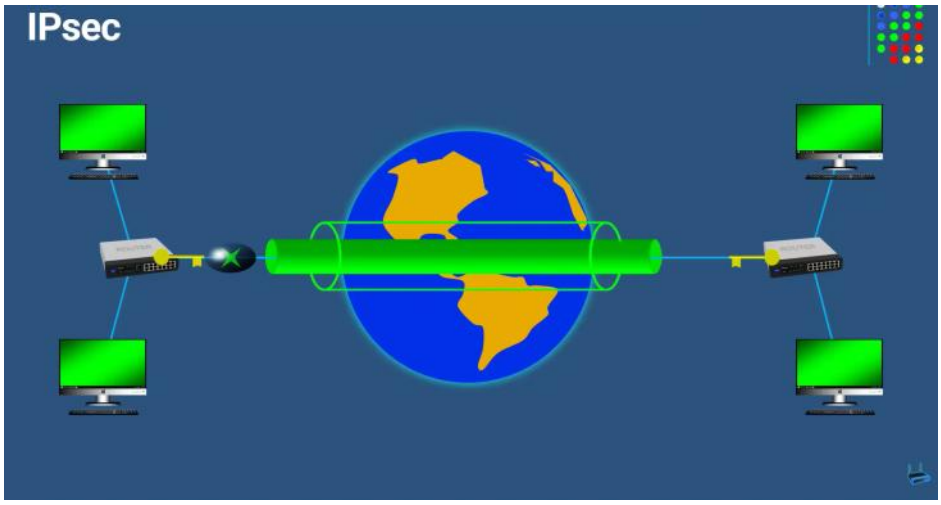
How it Works:





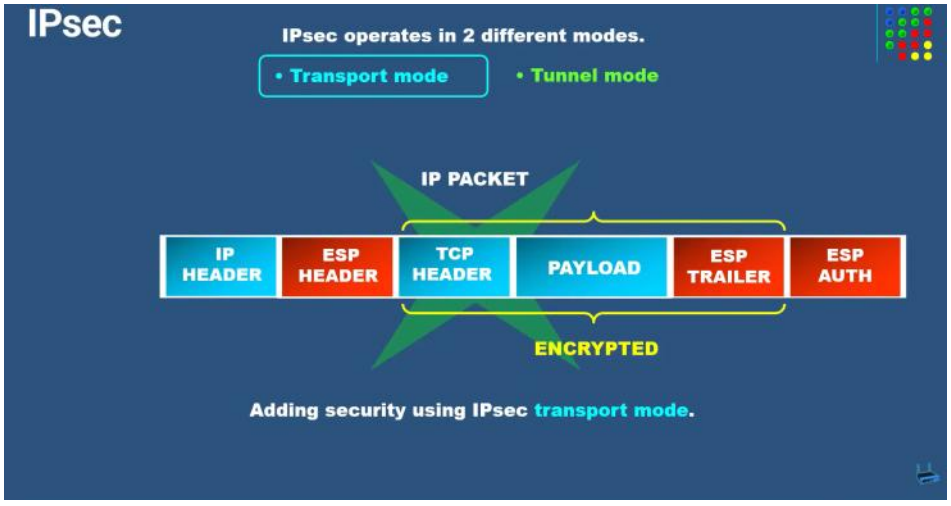
Internet Key Exchange (IKE): Operates in two phases to first authenticate the devices and negotiate encryption parameters, and second to establish the secure tunnel for data

exchange.
Encryption Protocols:
AH (Authentication Header): Provides authentication and integrity but lacks encryption.
ESP (Encapsulating Security Payload): The preferred protocol, offering authentication, integrity, and encryption to scramble data against theft.

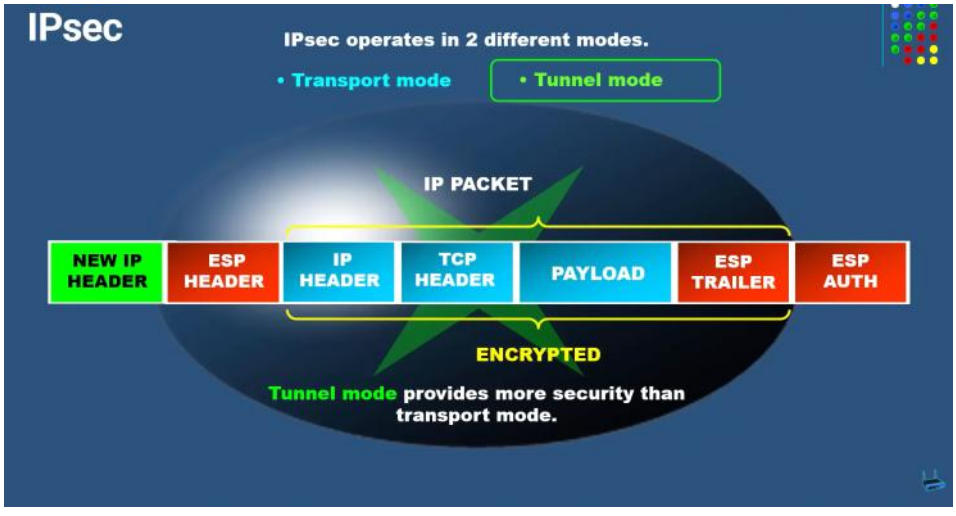


Key Management: Sender and receiver share keys to lock (encrypt) and unlock (decrypt) data as it travels across public networks.

Operating Modes:



Transport Mode: Used for device-to-device communication; encrypts the payload and headers but keeps the original IP header visible.



Tunnel Mode: Used for site-to-site VPNs; encrypts the entire original packet and adds a new IP header, providing higher security.